HAJU PAINTS SOLUTIONS

Mobile App Development Proposal

Executive Summary

HAJU Paints Solutions aims to revolutionize the paint industry in Kenya by developing a comprehensive digital platform that connects customers with high-quality paints and professional painting services. This proposal outlines the development of a mobile application ecosystem comprising a customer mobile app, painter registration portal, and admin management system.

1. Project Overview

Project Name: HAJU PAINTS SOLUTIONS APP

Project Duration: 3 months

Project Type: Mobile App Development with Web Admin Portal

Primary Objectives:

- Enable customers to purchase paints and book painting services remotely
- Create a platform for painters to register and manage their availability
- Implement seamless payment integration with M-Pesa
- Provide comprehensive customer support and feedback mechanisms
- Establish an efficient admin portal for business management

2. System Architecture

The HAJU Paints Solutions platform will consist of three main components:

2.1 Customer Mobile App (Angular Ionic)

- Cross-platform mobile application for iOS and Android
- User-friendly interface for paint selection and service booking
- Real-time notifications and order tracking
- Integrated payment gateway with M-Pesa

2.2 Admin Portal (Angular)

- Web-based administration dashboard
- User management and analytics
- Order processing and painter assignment
- Inventory management and reporting

2.3 Backend API System (Go)

- RESTful API architecture
- Database management with MySQL
- Third-party integrations (M-Pesa, Africastalking)
- Security and authentication protocols

3. Core Features

3.1 Customer Features

- **Paint Catalog:** Browse and purchase interior, exterior, and industrial paints
- **Service Booking:** Schedule professional painting services
- **Color Consultation:** Virtual color matching and recommendations
- **Order Tracking:** Real-time updates on paint delivery and service appointments
- **Payment Integration:** Secure M-Pesa payment processing
- **Customer Support:** Live chat, complaints, and feedback system
- **Order History:** Track past purchases and services

3.2 Painter Features

- **Registration Portal:** Professional painter onboarding
- **Availability Management:** Real-time status updates (available/engaged)
- **Job Assignments:** Receive and accept service requests
- **Earnings Tracking:** Monitor completed jobs and payments
- **Profile Management:** Update skills, certifications, and portfolio

3.3 Admin Features

- **Dashboard Analytics:** Sales, orders, and performance metrics
- **User Management:** Customer and painter account administration
- **Inventory Control:** Paint stock management and procurement
- **Order Processing:** Manage paint orders and service requests
- **Payment Management:** Track transactions and painter payments
- **Reporting:** Generate business intelligence reports

4. Technical Specifications

4.1 Frontend Technologies

- **Customer Mobile App:** Angular Ionic Framework
    - Cross-platform compatibility (iOS/Android)
    - Native device features access
    - Push notification support
- **Admin Portal:** Angular Framework
    - Responsive web design
    - Material Design components
    - Real-time data updates
    - Advanced filtering and search

4.2 Backend Technologies

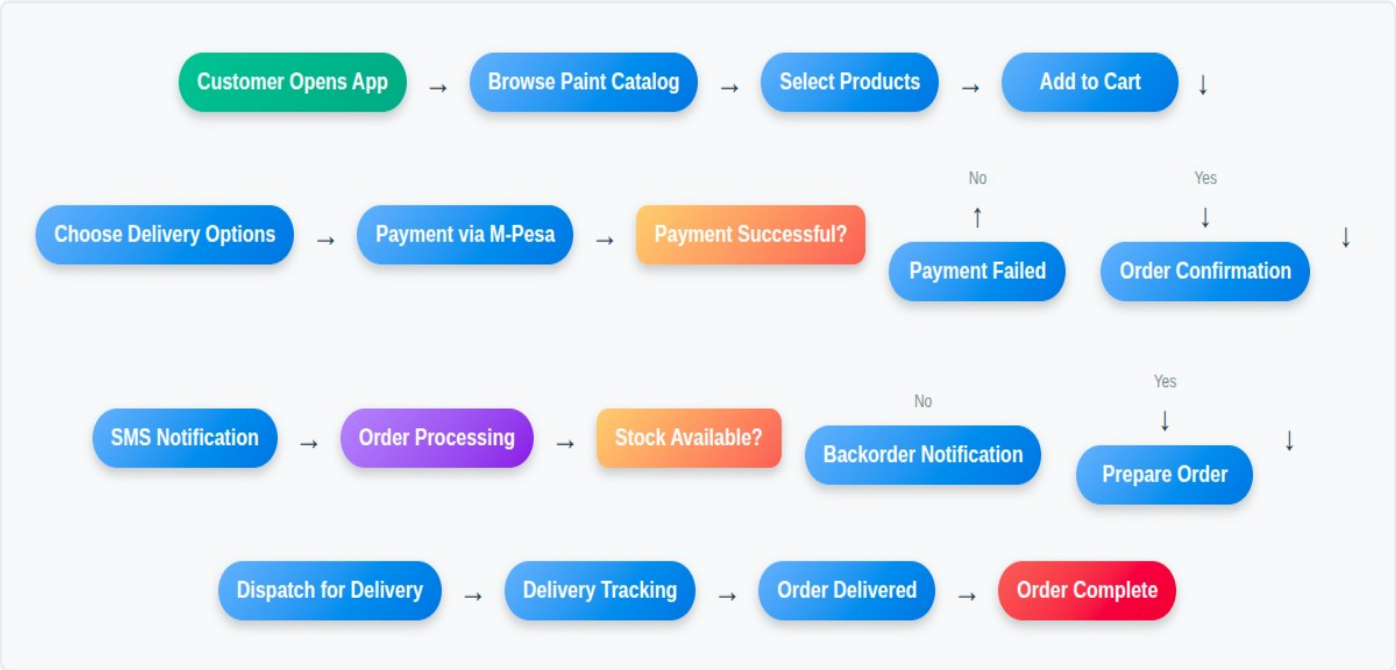
- **Server-Side:** Go Programming Language
    - High-performance HTTP server
    - Concurrent request handling
    - RESTful API architecture
    - JWT authentication and authorization
- **Database:** MySQL
    - Relational database management
    - Data integrity and consistency
    - Optimized query performance
    - Backup and recovery protocols

4.3 Third-Party Integrations

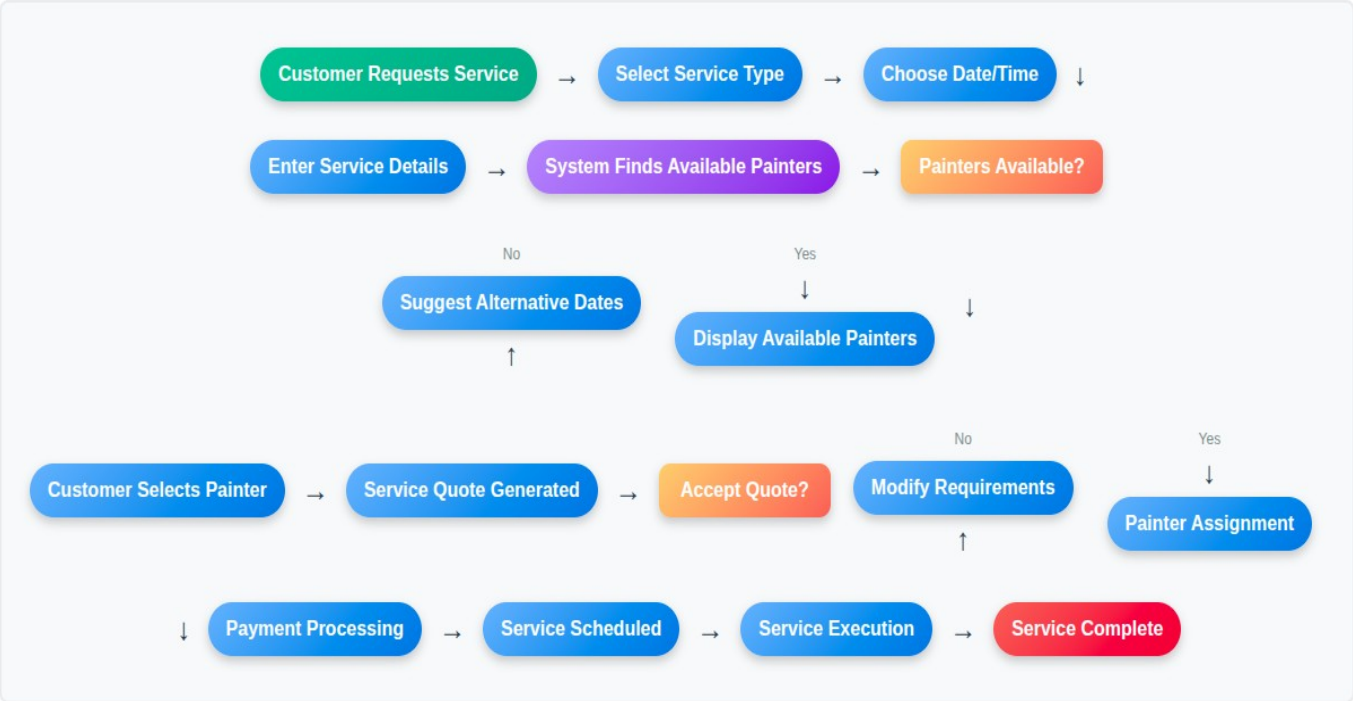
- **M-Pesa API:** Payment processing and transaction management
- **Africastalking SMS API:** Automated notifications and alerts.
- **Email Integration :** Notification and alerts

5. System Flow Diagrams

5.1 Customer Order Flow



5.2 Service Booking Flow



5.3 Painter Registration Flow



5.4 Admin Management Flow

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Admin Dashboard → Monitor Orders → Manage Inventory →
Process Payments → Assign Painters → Generate Reports →
Customer Support → System Maintenance
```

6. Development Timeline

Phase 1: Planning & Design (2 weeks)

- Requirements analysis and documentation
- UI/UX design for mobile app and admin portal
- Database schema design
- API specification documentation
- Project setup and development environment

Phase 2: Backend Development (4 weeks)

- Database implementation and configuration
- RESTful API development
- Authentication and authorization system
- M-Pesa payment integration
- SMS notification system
- Testing and documentation

Phase 3: Frontend Development (4 weeks)

- Customer mobile app development
- Admin portal development
- API integration and testing
- UI/UX implementation
- Cross-platform compatibility testing

Phase 4: Integration & Testing (1 week)

- System integration testing
- Security testing and vulnerability assessment
- Performance optimization
- User acceptance testing
- Bug fixes and improvements

Phase 5: Deployment & Launch (1 week)

- Production environment setup
- App store submission
- Admin portal deployment
- User training and documentation
- Go-live support

7. Budget Estimate

7.1 Development Costs

Component	Cost (KSH)
Backend Development (Go + MySQL)	35,000
Customer Mobile App (Angular Ionic)	35,000
Admin Portal (Angular)	45,000
Third-party Integrations (Mpesa, Bulk SMS and Email)	10,000
Testing & Quality Assurance	0
Project Management	0
Subtotal	**125,000**

7.2 Infrastructure & Licensing (The customer to be responsible for renewal)

Component	Cost (KSH)
Cloud Hosting (Renewed Annually)	14,000
App Store Fees (Paid once to google)	3,250
SMS Sender ID (Paid Once)	8,000
Subtotal	**25,250**

7.3 System Maintenance Fee

To pay 20% of the total development cost per year after Go Live date. This will facilitate 24 hrs technical support. The customer can opt not to be supported.

8. Risk Assessment & Mitigation

8.1 Technical Risks

- **Risk:** Third-party API limitations or downtime
- **Mitigation:** Implement fallback mechanisms and error handling
- **Risk:** Security vulnerabilities
- **Mitigation:** Regular security audits and penetration testing

8.2 Business Risks

- **Risk:** Low user adoption
- **Mitigation:** Comprehensive marketing strategy and user incentives
- **Risk:** Painter onboarding challenges
- **Mitigation:** Streamlined registration process and attractive commission structure

9. Post-Launch Support

9.1 Maintenance & Updates

- Regular bug fixes and performance improvements
- Feature enhancements based on user feedback
- Security updates and patches
- Third-party integration maintenance

9.2 Support Services

- 24/7 technical support
- User training and onboarding
- System monitoring and maintenance
- Regular backup and disaster recovery

10. Conclusion

The HAJU Paints Solutions mobile app platform represents a significant opportunity to digitize and streamline the paint industry in Kenya. With a comprehensive feature set, robust technical architecture, and strategic implementation plan, this project will position HAJU Paints as a market leader in digital paint retail and services.

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